

03/03/25

## Lecture 16

### Ordering and Broken Symmetry

- Systems with only short range interaction in 1D do not order at any non-zero temperature according to the exact solution obtained by the transfer matrix method for 1D Ising nearby spin interaction model.
- Mean field theory works good for higher than 1 dimensions & it is exact for  $d=4$  & higher.
- Peierls proof was rigorous which proved order in 2D Ising model with only nearby interactions at low enough temperature.

- The ordered state usually breaks a symmetry of Hamiltonian.

Eg:-

(i) Ising model hamiltonian has discrete symmetry called  $\mathbb{Z}_2$  or up-down symmetry, where if every spin is reversed, even then the hamiltonian is invariant.

$$H = \frac{1}{2} \sum_{i,j} J_{ij} S_i S_j$$

If all spin  $S_i \rightarrow -S_i$ , still the hamiltonian remains the same.

But an ordered state, say the ground state is not invariant under all  $S_i \rightarrow -S_i$ .

$|\uparrow\uparrow\uparrow\uparrow\cdots\uparrow\rangle \xrightarrow{S_i \rightarrow -S_i} |\downarrow\downarrow\downarrow\downarrow\cdots\downarrow\rangle$   
Ground state of ising model                      This is a different degenerate ground state.

Thus the ground state doesn't respect the symmetry of hamiltonian. This is called as Spontaneous Symmetry breaking as at low temperature ( $T=0$ ), the system spontaneously orders even when external  $h=0$ .

This is actually the breaking of time reversal symmetry. As it is the time reversal transformation which makes the spins to flip.

(ii) In Heisenberg ferromagnet,

$$\hat{H} = -J \sum_{\langle i,j \rangle} \hat{\vec{S}}_i \cdot \hat{\vec{S}}_j$$

This hamiltonian is invariant under rotation in the spin space as it is the spin operators which will generate rotations.

The rotation is  $\hat{R}(\vec{n}, \theta) = e^{-i\theta \hat{\vec{S}} \cdot \vec{n}}$

$\vec{n} \Rightarrow$  direction of rotation axis.

$$[\hat{H}, \hat{R}(\vec{n}, \theta)] = \sum_{\langle i, j \rangle} [\hat{\vec{S}}_i \cdot \hat{\vec{S}}_j, e^{-i\theta \hat{\vec{S}} \cdot \vec{n}}]$$

where  $\vec{S} \Rightarrow$  total spin operator  
all constituent spin operators  
will commute with total spin.

$$\hat{\vec{S}} = \sum_j \hat{\vec{S}}_j, \quad [\hat{S}_i, \hat{S}_j] = 0$$

$$\Rightarrow e^{-i\theta \hat{\vec{S}} \cdot \vec{n}} = \prod_j e^{-i\theta \hat{\vec{S}}_j \cdot \vec{n}}$$

$$\Rightarrow [\hat{H}, \hat{R}(\vec{n}, \theta)] = \sum_{\langle i, j \rangle} [\hat{\vec{S}}_i \cdot \hat{\vec{S}}_j, \underbrace{\prod_j e^{-i\theta \hat{\vec{S}}_j \cdot \vec{n}}}_0]$$

Thus the Heisenberg ferromagnet hamiltonian has the rotational invariance. This is a continuous symmetry possessed by Heisenberg. But the ground state will be such that all spins point in some preferential direction. Thus here a continuous symmetry is broken.

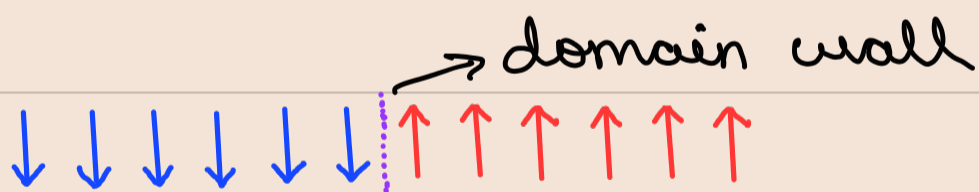
(iii) In Crystals, the full microscopic hamiltonian which describes the interaction of particles in a periodic potential, has continuous translation symmetry, i.e. it doesn't depend on the choice of coordinate system, it can be translated arbitrarily small or large. But the crystal state chooses to be in a periodic structure spontaneously & the crystal's ground state is not having continuous symmetry rather it has discrete translational symmetry.

- When an ordered state of the Hamiltonian breaks a continuous symmetry, then there can exist low lying excited states which are arbitrarily close to the ground state.

These modes are called as Goldstone modes.



- In low dimensions the thermodynamic consequences can be drastic. There the thermal occupation of low lying states can destroy a putative long range order (Eg: London's argument)
- In discrete Spin One dimensional Ising model with only nearby spin interaction, we can get an excited state with anti periodic boundary conditions, as shown below.



The energy cost is,  $\Delta E = 2J$ .

Where as if all spins are aligned in same direction, then  $E = -NJ$ .

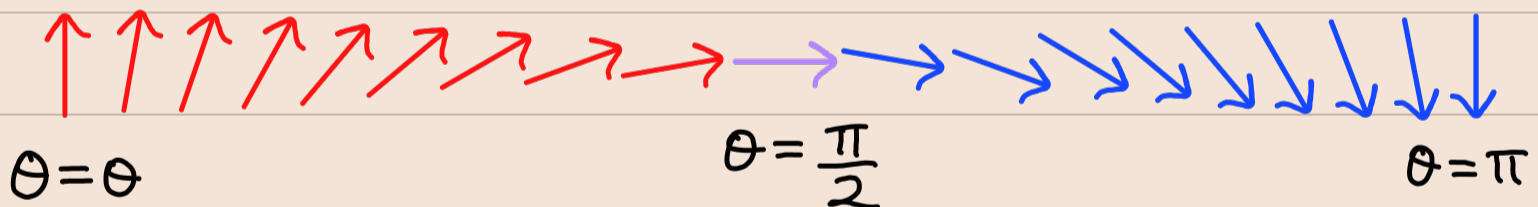
Thus here discrete symmetry is broken. Hence there is a proper separation b/w ground state & the excited states.

- In 1D continuous Spins model, the Hamiltonian is given by,

$$H = -J \sum_{\langle i,j \rangle} \vec{S}_i \cdot \vec{S}_j$$

$$\hat{H} = -J \sum_{\langle i,j \rangle} \cos(\theta_{ij})$$

where  $\theta_{ij} \Rightarrow$  angle b/w neighbouring spins. With anti-periodic boundary condition, the system state will be as follows,



$N$  spins spread over a distance ' $L$ '.  
This state has neighbouring spins deviated by an angle of ' $\frac{\pi a}{L}$ '

$$\Rightarrow \theta_{i,i+1} = \frac{\pi a}{L} \text{ for all } i, j.$$

$a \Rightarrow$  lattice spacing

$$E = -J \sum_{\langle i, j \rangle} \cos(\theta_{i, i+1})$$

$$= -J \sum_{\langle i, j \rangle} \cos\left(\frac{\pi a}{L}\right)$$

$$E = -NJ \cos\left(\frac{\pi a}{L}\right)$$

$$E = -NJ \left[ 1 - \frac{1}{2!} \left(\frac{\pi a}{L}\right)^2 + \frac{1}{4!} \left(\frac{\pi a}{L}\right)^4 - \dots \right]$$

$$\Rightarrow E - E_{g.s.} = NJ \left[ \frac{1}{2!} \left(\frac{\pi a}{L}\right)^2 - \frac{1}{4!} \left(\frac{\pi a}{L}\right)^4 + \dots \right]$$

$$E_{g.s.} = -NJ$$

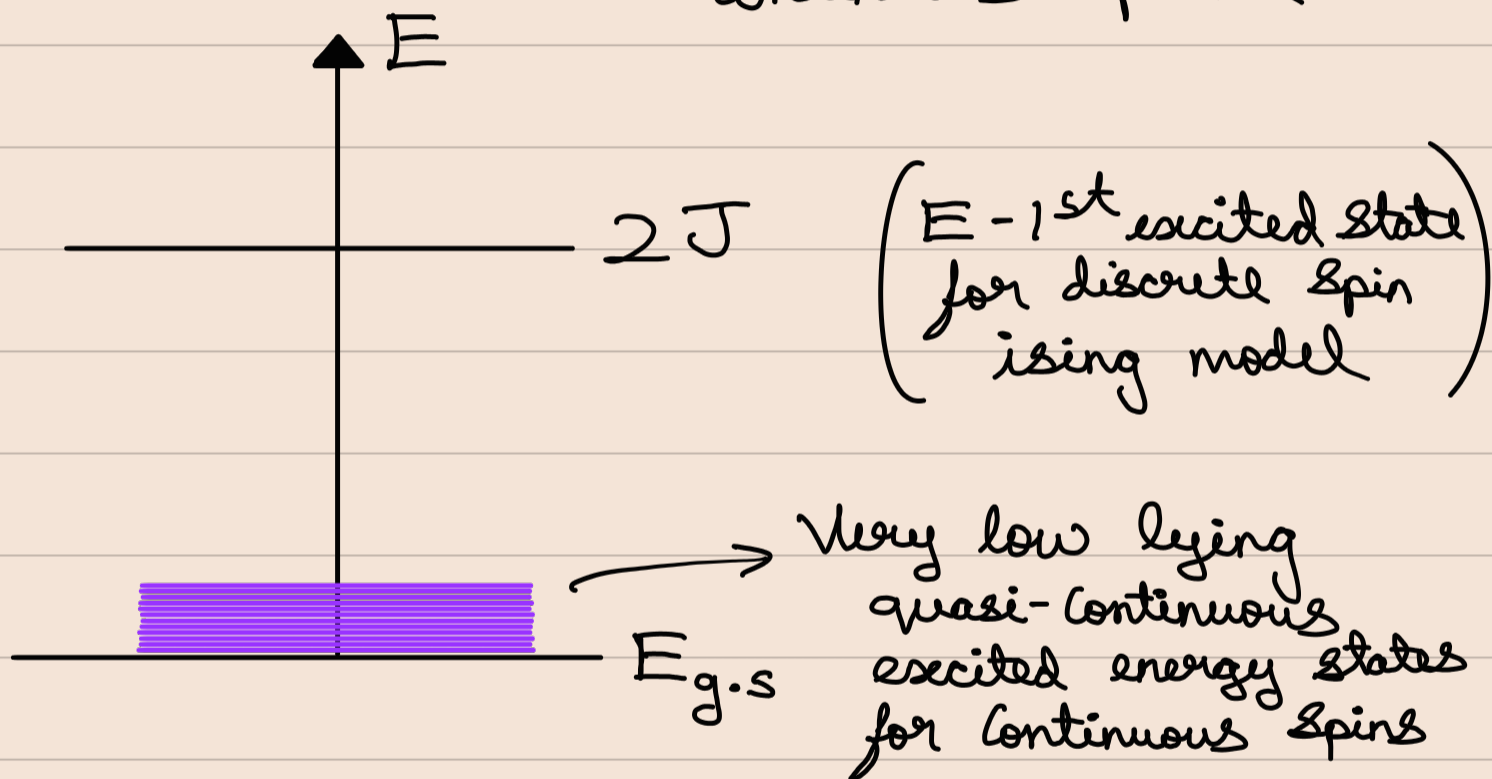
$$\Rightarrow E_{\text{excited}} - E_{g.s.} \sim \frac{NJ}{2!} \left(\frac{\pi a}{L}\right)^2$$

$$\Rightarrow \Delta E \sim \frac{NJ \pi^2 a^2}{2L^2}$$

$$\text{Now } N = \frac{L}{a} \quad \Rightarrow \quad \Delta E = \frac{\pi^2 J a}{2L}$$



Thus in thermodynamic limit,  $N \rightarrow \infty$  &  $L \rightarrow \infty$ . Thus  $\Delta E$  is very negligible. Thus there can be low lying states very close to the ground state which have almost continuous spectra.



Thus it is very easy to destroy order in the case of Continuous Spin whose hamiltonian has a Continuous Symmetry in it.